

WASP-35b

R-1024

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_251226 · created 2026-07-30T05:07:37+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461035.7709 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.01
Transit depth (Rp/Rs)^2	2.34 +/- 0.32 [%]
Orbital Inclination (inc)	87.53 +/- 0.96
Airmass coefficient 1 (a1)	1.2609 +/- 0.009
Airmass coefficient 2 (a2)	-0.146 +/- 0.013
Scatter in the residuals of the lightcurve fit is	0.72 %
Best Comparison Star	None
Optimal Aperture	4.34
Optimal Annulus	10.58
Transit Duration (day)	0.1316 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.01 vs 0.1241 (deviation 1.97σ)
Duration fit vs published	0.1316 d vs 0.1317 d (deviation 0.01σ)
Mid-time O–C	0.2 min
Depth SNR	10.4
Residual scatter	0.72 %

Light curve

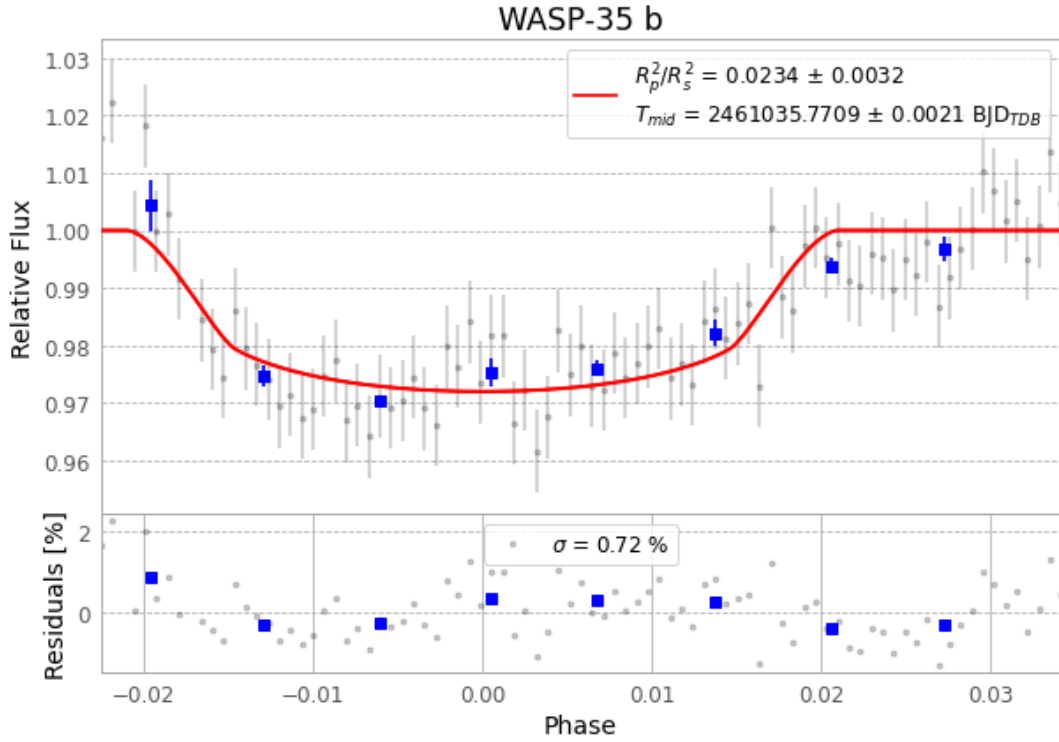


Figure 1 — FinalLightCurve_WASP-35 b_2025-12-25.png (SHA-256 77b43840ccb5cd05...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [358, 293], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 947b8f77c93d5e7f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit f8439d0

Data provenance

87 FITS frames (57 MB), WASP-35251226043915.FITS → WASP-35251226085715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-251226.log (50f8fdbde539...), FinalLightCurve_WASP-35 b_2025-12-25.png (77b43840ccb5...), FinalLightCurve_WASP-35 b_2025-12-25.csv (43ad9ece0556...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 05:07Z.